

Reading the Fab from the Machine Room: Recovering Per-Die Silicon Harvest Maps from Supercomputer Out-of-Band Telemetry

Mohsen Seyedkazemi Ardebili

Affiliation

mohsen.seyedkazemi@gmail.com

[TODO: co-authors]

Abstract—Processor vendors sell partially defective dies by permanently disabling faulty units, a practice known as core harvesting. The resulting per-die disable maps are proprietary and are conventionally invisible to system owners. We show that the out-of-band (BMC/IPMI) sensor telemetry routinely collected by production supercomputers exposes these maps directly, and we recover them for all 1,960 processor sockets of CINECA’s Marconi100 from the public M100 ExaData release [1]. Every socket exposes exactly 16 of 24 possible per-core thermal sensors, and the eight absent sensors always form four complete *slice* pairs $(2k, 2k+1)$ —the POWER9 unit that shares an L2/L3 block [2], [3]—with zero exceptions. We observe 442 of the 495 combinatorially possible patterns, establishing that the map is per-die rather than a fixed SKU. Using a curveball null that preserves both per-socket and per-slice marginals, we show harvested slices cluster ($z = -20.2$), and a fitted pairwise-interaction model separates that clustering from per-slice propensity, showing both are present and that the two procurement lots differ in each. Thermal analysis over 240 sockets independently recovers the sensor-index ordering as the physical arrangement of slices on the die (linear fit $z = -6.13$; an unconstrained search returns the same ordering reversed, $|\rho| = 0.993$), so the clustering is spatial rather than merely index-space, and confirms that the two cores of a slice are physically abutted ($r = +0.41$ within-slice versus $+0.32$ across a slice boundary at identical index distance). We detect a sharp procurement-lot boundary at rack 22 (Welch $t = 27.3$) separating two populations with markedly different harvesting statistics. Finally, tracking every socket across the full 2.5-year, 1.67-million-socket-day record, we find harvest maps behave as near-static manufacturing constants: 89.5% of sockets never change configuration, and none of the 235 observed transitions occurs during operation—every one coincides with a reboot. Against that backdrop we isolate three episodes, on three of 1,968 sockets, in which a socket ran on 14 cores for 2–4 days, each losing exactly one slice and at least one subsequently restored: to our knowledge the first direct observation of POWER9 field core-guarding in a production system, at a measured rate of 0.15% of sockets over 2.5 years.

Index Terms—core harvesting, silicon binning, HPC telemetry, POWER9, yield, hardware provenance

I. INTRODUCTION

Every large processor die ships with defects. Rather than discard them, vendors permanently fuse off the faulty units and sell the remainder as a lower-core-count part. We call this *silicon harvesting*, or *die harvesting*, to distinguish it from the now more common architectural sense of “core harvesting” as the reclamation of idle CPU *cycles* for co-located work [4];

the two are unrelated. The practice is universal and openly acknowledged, but the resulting *per-die* maps of which units were disabled are proprietary: a buyer is told the part has 16 cores, never *which* 16. Those maps encode information about a fab’s defect process, a vendor’s binning policy and a procurement’s provenance that is otherwise unavailable outside the manufacturer.

Independently, HPC operators have begun publishing holistic out-of-band monitoring at fleet scale [1]. We observe that such telemetry carries an unintended side channel: a baseboard management controller exposes one thermal sensor per *physical* core position and does not renumber the surviving cores compactly, so the set of sensors that report *is* the die’s harvest map, published inadvertently and at fleet scale.

We exploit this to recover harvest maps for every processor socket of CINECA’s Marconi100, a Top-10 system, from a public dataset, and to track them across 858 days. Our contributions are:

- A recovery method (Sec. III) with an explicit validity argument, needing only per-core sensor telemetry that operators already collect.
- The first fleet-scale characterisation of core harvesting in a deployed supercomputer: the granularity is exactly the POWER9 slice, maps are per-die rather than per-SKU, and harvested slices cluster in sensor-index space.
- Evidence of *procurement-lot structure*—a sharp change-point separating two populations with markedly different binning statistics—from telemetry alone.
- A longitudinal result over 1.67M socket-days: maps are static, configuration changes occur only at reboot, and three field core-guard episodes are isolated and corroborated.
- A negative result that matters operationally: the harvest map has no practical effect on socket power (Sec. IV-J), so schedulers need not model it.

II. BACKGROUND

A. POWER9 slice organisation

The POWER9 die used in the IBM Power System AC922 contains 24 SMT4 cores (equivalently 12 SMT8 cores) on a 695 mm^2 die in 14 nm FinFET with 17 metal layers [3].

Critically, *each pair of SMT4 cores—or one singleton SMT8 core—comprises a slice, and each slice owns 512 kB of L2 and 10 MB of L3* [3], [2]. IBM’s Hot Chips 28 presentation shows the 24 cores above 12×10 MB L3 blocks, attached to a 7 TB/s on-chip fabric at 256 GB/s $\times 12$ [2]. The slice is therefore the natural fusing granularity: disabling one SMT4 core would strand its half of a shared L2/L3 block.

Marconi100 nodes are AC922 model 8335-GTG with 2×16 -core POWER9 at 2.6/3.1 GHz and $4 \times$ NVIDIA V100 [5], [6]. **A 16-core part is a 24-core die with four slices fused off**; this is the object of our study. Fig. 1 shows the organisation annotated with the harvest rates we measure.

B. The M100 ExaData dataset

M100 ExaData [1] publishes 49.9 TB of telemetry from Marconi100 covering 2020-03 to 2022-09 across 980+ nodes. The IPMI plugin collects BMC sensor data at 20 s cadence. The documentation describes the core-temperature family as `pX_coreY_temp`, “Temperature of core n. Y in the CPU socket n. X. X=0..1, Y=0..23” [7]. *This specifies the BMC sensor-ID namespace, not the active core count*—the distinction is the entry point for this study.

III. METHOD

A. Recovering the disable map

For each month we enumerate the `metric=pX_coreY_temp` partitions and count samples per (node, socket, core). A core index is *present* for a socket if it emits any sample in the window. We define the disable map of socket s as $M[s, k] \in \{0, 1\}^{12}$ over slices $k = \lfloor \text{core}/2 \rfloor$.

The key validity argument is that **the BMC does not renumber active cores compactly**: were that so, every socket would report indices 0–15. Instead we observe arbitrary 16-element subsets of $\{0, \dots, 23\}$, so the index is a fixed physical-position identifier.

B. Null model for clustering

Every socket has exactly four harvested slices, so the constraint induces negative dependence and a naive independence null is invalid. We use the curveball algorithm [8], which preserves *both* row sums (4 per socket) and column sums (per-slice marginals) while randomising association structure. Row- and column-sum preservation is asserted on each of 200 draws.

C. Thermal probing of die topology

Cores on a socket share workload and inlet air, producing a dominant common mode. We remove it by subtracting the per-timestamp cross-core mean, double-centre, and compute residual correlations. Physically abutted cores should retain positive residual coupling.

Caveat: Common-mode removal forces each row of the residual matrix to sum to approximately zero, so negative values at large index gaps are partly mechanical. We therefore rely on contrasts *at identical index distance*, which control for this exactly.

IV. RESULTS

A. Every socket is 16 cores; granularity is the slice

Populations: Three socket populations recur and are kept distinct throughout: $\mathcal{S}_{\text{all}}$, the 1,968 sockets appearing anywhere in the record (984 nodes); $\mathcal{S}_{\text{map}}$, the 1,962 with at least one clean day and hence a well-defined harvest map; and $\mathcal{S}_{2004}$, the 1,960 present in the single month 2020-04. All structural results below are over $\mathcal{S}_{\text{map}}$ and the full 31-month record.

TABLE I
DISABLE-MAP STRUCTURE OVER THE FULL 31-MONTH RECORD.

Sockets with a well-defined harvest map	1,962
Median clean days supporting each map	775
Socket-days reporting exactly 16 cores	99.985%
Sockets ever reporting 24 cores	0
Harvested cores form complete $(2k, 2k+1)$ pairs	1,962 (100%)

The perfect pair structure is the central mechanical finding: it matches the POWER9 slice definition exactly and rules out per-core fusing.

B. The map is per-die, not per-SKU

443 distinct patterns appear among the $\binom{12}{4} = 495$ possible (Fig. 7). The most common (slices 0–3) covers only 5.5% of sockets. Within a node the two sockets match in just 1.2% of cases, with mean overlap 1.648 slices against a shuffled null of 1.474 ± 0.026 ($z = +6.6$)—nearly independent, with a small excess attributable to same-lot pairing.

C. Marginals are strongly non-uniform

Slice 0 is harvested on 61.2% of sockets against a uniform expectation of 33.3% ($\chi^2 = 807.2$, $\text{df} = 11$, $p \ll 0.001$); Fig. 2 splits this by lot. These marginals are remarkably stable in time: across the 31 months the largest standard deviation of any slice’s monthly rate is **0.25 percentage points**, and slice 0 ranges only 60.9–61.8% (Fig. 8). The single-month snapshot used in earlier drafts was therefore representative, and every estimate here is a 2.5-year-stable measurement.

D. Harvested slices cluster in index space

Under the curveball null the mean index gap between harvested slices is 4.242 versus 4.582 ± 0.017 ($z = -20.2$). Applying a Bonferroni correction across all $\binom{12}{2} = 66$ pair tests ($|z| > 3.83$), 17 pairs remain significant (Fig. 3). The positive excesses are 0&1 ($z = +16.7$), 6&7 (+9.2), 2&3 (+8.0), 8&9 (+7.7), 10&11 (+7.5), 4&5 (+7.1), 1&2 (+6.9) and 0&2 (+6.3): **eight of the nine positive-excess pairs have $|\Delta k| \leq 2$** , and every significant *deficit* is a distant pair. We report the single exception rather than rounding it away. Because Sec. IV-F shows index adjacency *is* physical adjacency, this is spatial clustering, not merely index-space clustering.

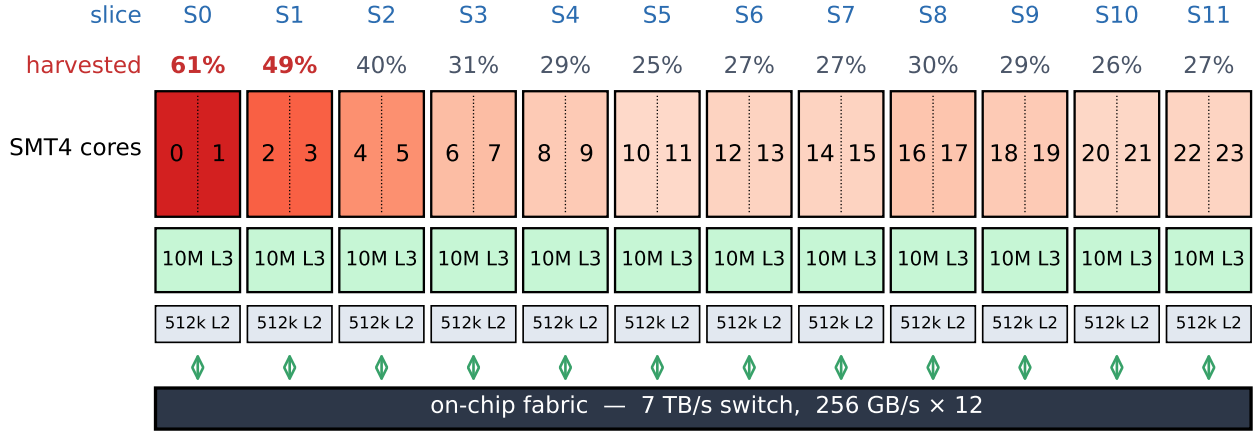


Fig. 1. POWER9 slice organisation (after [2]) annotated with the measured per-slice harvest rate across 1,960 Marconi100 sockets. Each slice holds two SMT4 cores sharing 512kB L2 and 10MB L3 on a common fabric port. Harvesting always removes whole slices. **The horizontal ordering is the IPMI sensor index, which Sec. IV-F shows is the physical die ordering, recovered independently from thermal coupling.**

E. Slice siblings are physically abutted

We probe 240 sockets—120 nodes, *both* p0 and p1—across the full 31-day month 2022-08, aligning all core streams on a common one-minute grid (Fig. 4). At *identical* index distance of 1, within-slice siblings reach $r = +0.407$ (95% CI [0.396, 0.419]) against $+0.318$ ([0.299, 0.337]) for cross-slice neighbours; the $+0.089$ gap has permutation $p < 10^{-4}$ and the intervals do not overlap. It replicates independently on each socket (p0: $+0.462$ vs $+0.336$; p1: $+0.353$ vs $+0.300$). The two cores of a slice are measurably more coupled than equally-numbered cores across a slice boundary, confirming physical abutment and the slice as a physical unit.

A correction to an earlier estimate: An initial version of this analysis used 40 sockets in 2020-04, a month with only three reporting days, and reported a contrast of $+0.51$ vs $+0.29$. That overstated the effect roughly threefold. The direction, the significance and the replication across sockets survive; the magnitude does not. We report the full-month estimate.

F. The sensor index recovers the physical slice ordering

Is the sensor index merely a label, or does it track physical position? We answer it by scoring explicit candidate floorplans: for each, correlate the predicted physical distance between two slices against their measured thermal coupling (Fig. 5). Coupling should fall with distance, so a better floorplan gives a more negative correlation.

The identity 1×12 linear arrangement wins decisively, at -0.754 against a random-ordering null of $+0.001 \pm 0.123$ ($z = -6.13$, $p < 10^{-4}$), and beats every two-dimensional alternative tried (6×2 : -0.716 ; 4×3 : -0.555 ; 3×4 : -0.488 ; 2×6 : -0.213). More stringently, an unconstrained hill-climbing search over one-dimensional orderings—free to return any of the $12!/2$ arrangements—converges to [10, 11, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0] at -0.759 , barely better than

the identity’s -0.754 . That is the sensor ordering *reversed*; since distance is invariant under reflection it is the same physical arrangement, differing by one adjacent transposition. Rank correlation with the identity is $|\rho| = 0.993$.

The thermal data therefore recovers the sensor-index ordering independently, up to reflection. This licenses reading Sec. IV-D spatially: slices adjacent in index are adjacent on the die, so the co-harvest excess is genuine *spatial* clustering of the kind the yield literature describes [9], [10].

A correction to an earlier estimate: On the three-day 2020-04 sample this test was inconclusive—the linear ordering scored only -0.283 ($p = 0.023$) and was beaten by a 3×4 grid—and we previously declined to claim a physical ordering. The full-month, both-socket measurement reverses that conclusion. We report both so the reader can judge how much the earlier sample size mattered.

G. A procurement-lot boundary at node 440

Node IDs map to position as $\text{rack} = \lfloor \text{id}/20 \rfloor$ [7]. A changepoint scan maximises Welch t at rack 22 (node 440), $t = 26.6$ (Fig. 6).

TABLE II
TWO POPULATIONS EITHER SIDE OF THE LOT BOUNDARY.

Population	Sockets	$P(\text{slice } 0)$	Distinct patterns
Lot A (racks 0–21)	880	88.2%	230
Lot B (racks 22–48)	1,082	39.3%	416

Lot A marginals fall steeply with slice index; Lot B is nearly flat. Permutation tests confirm between-rack heterogeneity for slices 0, 1, 2, 5 ($p < 0.01$), and same-rack sockets share more harvested slices than chance (1.568 vs 1.454 ± 0.008 , $z = +14.1$).

H. Configuration lifetime

POWER9 exposes mechanisms that *could* alter the active set after manufacture: predictive Dynamic Processor Deallocation, persistent GARD records read by hostboot at IPL and clearable with `opal-gard`, and the field core override used for licensing [11], [12]. Operating-system core offlining is invisible to the BMC and cannot affect sensor presence. Whether harvest maps are static in practice is therefore an empirical question, which we answer by aggregating every core sample to (node, socket, day) counts across the dataset. All numbers in this section are over the complete 31-month record (2020-03 to 2022-09).

a) *The count is very nearly invariant*: Across all 31 months—1,671,297 socket-days over 858 days and 984 nodes—**99.985% report exactly 16 active cores**. Of the 256 exceptions, 39 have an odd count, which *cannot* arise from slice-granular fusing and therefore proves the collector can drop individual cores; the counts above 16 are unions spanning a same-day reconfiguration. Only **nine** socket-days are fully-sampled with an even count below 16, and these are the subject of the guard analysis below.

b) *Configurations are near-static*: Restricting to *clean* socket-days (16 cores, each at >90% of the 4,320 samples expected at 20s cadence) leaves 1,517,473 socket-days, a median of 775 observed days per socket. **89.5% of sockets never change configuration** (1,756 of 1,962); 183 have two distinct sets, 19 have three, three have four and one has five.

c) *Three core-guard episodes*: The nine anomalous socket-days resolve into **three sustained episodes on three distinct sockets**, each with the exact signature of a field deconfiguration (Table III). In every case the socket loses *exactly one slice*—a complete $(2k, 2k+1)$ pair—runs on 14 cores for 2–4 consecutive fully-sampled days, and in at least one case the slice is subsequently restored, consistent with an operator clearing the GARD record [12]. The measured rate is **3 episodes across 1,968 sockets over 2.5 years**, i.e. 0.15% of sockets. Durations are lower bounds, since an episode is counted only on fully-sampled days.

Ruling out a collector fault. The competing explanation is that the collector dropped exactly those two metric streams. We test it directly: for each episode we ask whether the *other* sensors on the same node—in particular on the same socket—kept reporting. They did. In all three episodes the two slice sensors read zero for the entire window while **all ten control metrics, including `pX_power` and `pX_vdd_temp` on the same socket, sustained a full 4,320 samples per day** (Fig. 9). A collection fault that silences precisely the two cores of one slice for several consecutive days, while leaving every other sensor on the same socket at full cadence, is not a plausible account. We therefore attribute these episodes to hardware deconfiguration; to our knowledge this is the first direct observation of POWER9 field core guarding in a deployed system.

d) *Changes never occur during operation*: Of **235** transitions across the full record, **zero** occur between consecutive reporting days: every one coincides with a gap in reporting

TABLE III

CORE-GUARD EPISODES. BASELINES ARE THE ADJACENT OBSERVED DAY, NOT THE SOCKET’S GLOBAL MODAL SET, SINCE TWO OF THESE SOCKETS ALSO CHANGE CONFIGURATION ELSEWHERE IN THEIR LIFE.

Socket	Dates	Days	Slice lost	Restored
347 p1	2020-03-13–16	4	4 (cores 8,9)	yes
411 p1	2021-11-13–15	3	6 (cores 12,13)	yes [†]
946 p0	2020-08-31–09-01	2	8 (cores 16,17)	not observed

[†]restored via a day that also spans a reconfiguration.

(median 2d), against a background in which only 4.2% of consecutive reporting days are more than one day apart. Configuration changes are boot-time events, exactly as the guard/deconfiguration mechanism predicts.

The only two apparent counterexamples confirm the rule once examined at full timestamp resolution. Nodes 315 (p1) and 342 (p0) appear to change within a single day, and produce the dataset’s only 20- and 22-core socket-days. In fact both cut over on 2021-03-02 across a *sub-day reporting gap*: on node 315 the outgoing cores {4, 5, 20, 21} report until 08:42 and the incoming cores {8, 9, 10, 11} resume at 09:10, a 28-minute outage, with exactly 16 cores before and 16 after; the 20/22-core counts are unions across the cutover. Node 342 cuts over in the same window, indicating a coordinated maintenance reboot. Day-granularity aggregation is therefore *conservative* here: it cannot see sub-day downtime, so our transition-gap statistics understate how consistently changes coincide with reboots.

e) *Changes concentrate on a few dates*: The 235 transitions involve 132 nodes on 81 distinct dates, but are heavily clustered: 2020-03-20 alone accounts for 63 and the top three dates for 41%. Such synchronised, fleet-wide changes are more consistent with an administrative or firmware event than with independent per-node hardware work (see OI-2).

f) *Some apparent changes are socket relabelling*: When node n ’s socket p0 loses exactly the slices p1 gains and vice versa, the dies are unchanged and only the p0/p1 labels have swapped. This accounts for **20%** of the 230 transitions where both sockets are observable.

g) *Caveat*: An apparent change can arise from (a) genuine hardware reconfiguration, (b) socket relabelling, or (c) a change in the BMC sensor-ID mapping, e.g. after a firmware update. The tight clustering of 86 transitions into three days is consistent with (c) and we cannot separate these from telemetry alone. **[TODO: seek CINECA maintenance logs for March 2020.]**

I. A clustering parameter, fitted per lot

Section IV-D establishes clustering against a null; it does not *quantify* it, and the yield models we invoke [9], [10] are parameterised by a clustering strength. Those models describe defect *counts* per die, which we cannot observe: the SKU fixes the count at exactly four harvested slices. What is observable is the conditional spatial arrangement given four, so we fit the

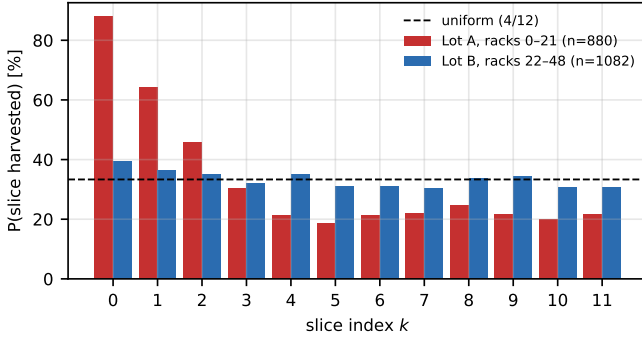


Fig. 2. Per-slice harvest probability by procurement lot.

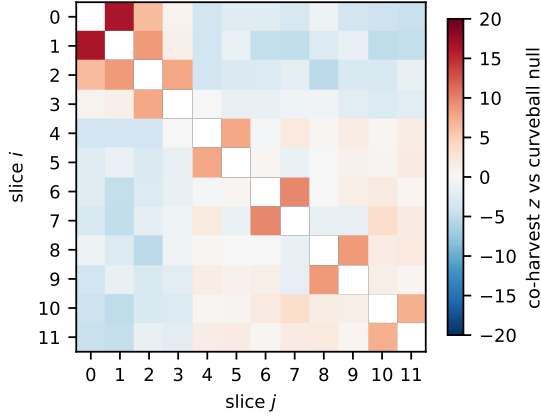


Fig. 3. Co-harvest z against the curveball null. Excess is confined to near-diagonal (index-adjacent) slice pairs.

natural conditional analogue—a pairwise-interaction model on the twelve slice positions,

$$P(S) \propto \exp\left(\sum_{k \in S} \beta_k + \gamma \text{adj}(S)\right),$$

where $\text{adj}(S)$ counts index-adjacent pairs in the harvested set S . The β_k absorb per-slice propensity (the slice-0 asymmetry) and γ measures clustering *beyond* what those propensities explain. The normalising constant is an exact sum over all $\binom{12}{4} = 495$ patterns, so likelihood and gradient are exact.

TABLE IV
FITTED CLUSTERING PARAMETER BY PROCUREMENT LOT.

Population	n	γ	LR χ^2_1 vs $\gamma=0$
Lot A (racks 0–21)	880	$+0.743 \pm 0.044$	174.2
Lot B (racks 22–48)	1,082	$+0.434 \pm 0.040$	106.4
Whole fleet	1,962	$+0.647$	394.5

Three things follow. First, **clustering is real in both lots** (χ^2_1 of 174 and 106, $p \ll 0.001$): mean adjacent-pair counts are 1.463 and 1.218 observed against 1.203 and 0.993 under the no-clustering model. Second, **the lots differ significantly in clustering strength** ($z = +5.22$), not only in their marginals. Third, and most usefully, the two effects *separate*: Lot A’s

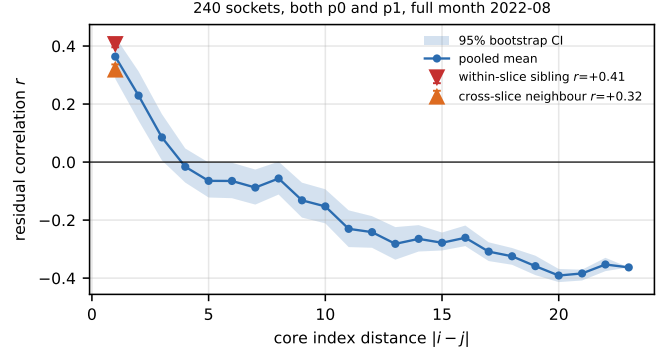


Fig. 4. Residual thermal correlation versus core index distance. The within-versus cross-slice contrast at distance 1 controls exactly for index gap.

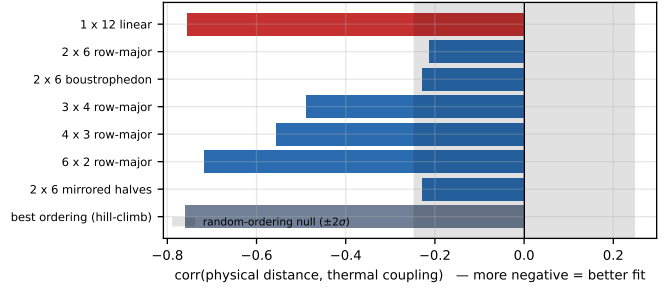


Fig. 5. Candidate floorplans scored against measured thermal coupling (more negative = better fit). The linear index ordering (red) wins decisively and is matched only by an unconstrained search that converges to the same arrangement reversed.

$\beta_0 = +2.51$ dwarfs every other propensity while Lot B’s β vector is nearly flat (range $+0.34$ to -0.13), yet both carry a substantial positive γ . The slice-0 asymmetry and the spatial clustering are therefore distinct phenomena rather than two views of one.

J. The harvest map has no operational consequence

Recovering a hidden hardware property is only useful if it changes something. We therefore ask whether the harvest map predicts socket behaviour, using a **within-node paired contrast**: the two sockets of a node share chassis, airflow, inlet air, power supply and—to first order—workload, so the difference $p_0 - p_1$ removes the node-level confounds that make cross-node power comparisons unreliable. We summarise each socket’s map by the mean index of its active slices, their dispersion, the number of adjacent active pairs, and the count of active slices in the low half, and correlate the socket-to-socket difference in each against the socket-to-socket difference in mean CPU power over six months (5,880 node-months, Fig. 11).

The associations are statistically detectable and practically negligible. The strongest is between the mean active-slice index and socket power: $r = +0.039$ (permutation $p = 0.003$), explaining **0.16% of variance**, with a slope of $+0.46$ W per unit of slice index (95% CI $[+0.16, +0.76]$). Across the

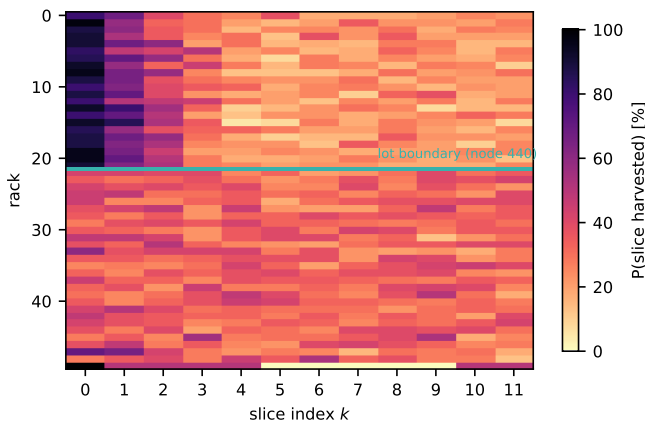


Fig. 6. Harvest probability by rack and slice. The lot boundary at rack 22 is visible as an abrupt change.

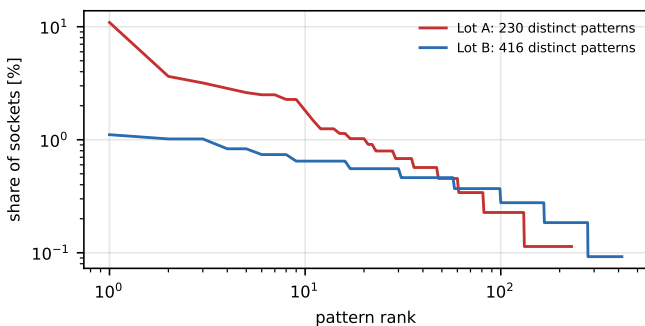


Fig. 7. Rank–frequency of disable patterns. Lot B is markedly more diverse.

full observed range of the predictor this is **3.1W**, against a socket-to-socket power spread of 17.8W. For calibration, the same paired design shows socket p0 draws 7.7W more than socket p1 on average ($t = 44.6$)—mere physical position in the chassis matters more than twice as much as the entire range of harvest-map variation.

We therefore report a deliberate negative result: **harvest maps are operationally invisible**. Schedulers, power models and node-ranking heuristics need not represent them, and node-to-node performance variability in this system must be sought elsewhere. The result also strengthens the provenance interpretation of Sec. IV-G: because the map has no behavioural footprint, the lot boundary cannot be an artefact of operational differences between racks.

V. DISCUSSION

a) Why is slice 0 special?: The single most striking number is that slice 0 is harvested on 88.2% of Lot A sockets but only 39.3% of Lot B. Two explanations are live. Under *defect-driven* harvesting, fusing follows wherever killer defects land. Under *policy-driven* harvesting, slice 0 abuts a shared structure and is preferentially fused for reasons of yield policy, binning recipe or physical design; an 88% rate on one specific slice is very hard to obtain from random defects.

The fit of Sec. IV-I shows the answer is *both*, and separates them. The propensity vector β carries the slice-0 asymmetry and differs enormously between lots ($\beta_0 = +2.51$ versus a flat Lot B); the interaction parameter γ carries the spatial clustering and is significantly positive in *both* lots. A defect process is operating throughout—consistent with the yield literature [9], [10]—while a strong additional slice-0 preference operates in Lot A only. The most economical reading is a change of binning recipe or die source between the two procurements, superimposed on a defect process present in both. We cannot settle the mechanism from one machine, but the decomposition is sharper than an either/or.

b) The direction of causation is fixed.: A tempting objection to Sec. IV-G is that rack position is confounded with cooling and workload. It cannot be: harvesting is performed by fusing at manufacture, months before a part is installed, so nothing about a socket’s position in the machine room can influence its harvest map. The only admissible causal direction runs from procurement batch to rack, which is precisely the provenance signal we claim.

c) What this buys an operator.: Three things. Fleet telemetry can audit a procurement without vendor cooperation, detecting whether a delivery is homogeneous. It can detect field deconfiguration events that would otherwise be visible only in service logs. And, per Sec. IV-J, it establishes when a hardware property can safely be *ignored*.

VI. RELATED WORK

All references in this section were verified against their original sources on 2026-08-05. Manufacturing variability in HPC has been studied mainly through its *performance* consequences—frequency and power spread under a power cap [13], turbo-boost variation across nominally identical processors [14], and accelerator-level variability in GPU-rich systems [15]. That literature treats the die as a black box characterised by its behaviour. Yield modelling, conversely, treats defect statistics directly: the negative-binomial family [9] and spatial wafer-map models [10] describe clustered killer defects but are validated on fab data unavailable to system owners.

A third, closer body of work concerns hardware provenance. Physical unclonable functions and inherent hardware identifiers exploit analog manufacturing variation to answer *which die is this*, for traceability and counterfeit detection [16]. That is a different question: we do not identify a die, we recover *what the vendor disabled on it*, without designed-in circuitry or physical access. Node-level anomaly and outlier detection in HPC [17], [18] likewise establishes *that* nodes differ without recovering the hardware structure responsible. The M100 ExaData release itself has an active downstream literature spanning telemetry foundation models, energy analytics and operational-data ontologies [19]; none of it examines per-core sensor presence.

We are not aware of prior work recovering per-die harvest maps from deployed-system telemetry, and that is the gap this paper addresses. We state this as an awareness claim rather

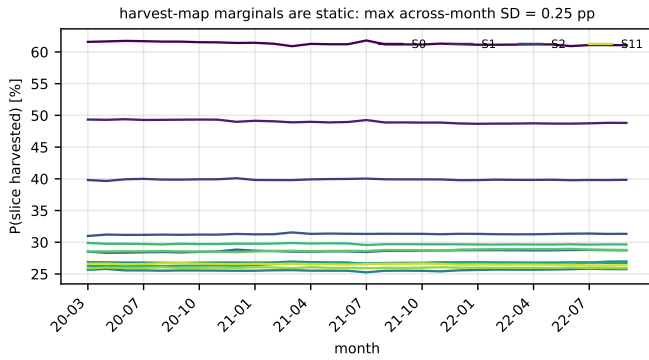


Fig. 8. Per-slice harvest rate in each of the 31 months. The largest across-month standard deviation of any slice is 0.25 percentage points.

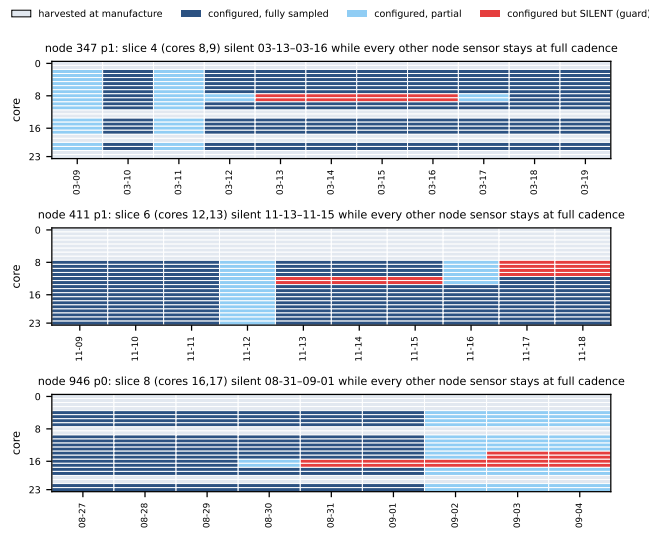


Fig. 9. The three core-guard episodes. Grey = harvested at manufacture, blue = configured and reporting, red = configured but silent. In every case exactly one slice goes silent while all ten control sensors on the same node stay at full cadence.

than a priority claim: it rests on a systematic but necessarily incomplete search.

VII. THREATS TO VALIDITY AND OPEN ISSUES

Every unresolved item is tracked with an identifier in `OPEN_ISSUES.md` in the artefact repository, which records what is known, what would settle it, and where to look. **The items marked *blocking* below must be resolved before this paper is submitted.**

- 1) **Sensor presence $\neq$ core existence.** Mitigated by 99.985% of 1.67 M socket-days reporting exactly 16 cores, agreement with the published 16-core SKU [5], the perfect slice-pair structure, and the control of Sec. IV-H showing that when core sensors do vanish, every other sensor on the socket keeps reporting.
- 2) **[OI-1] Resolved.** The sensor index is recovered as the physical slice ordering from thermal coupling (Sec. IV-F), so the spatial reading of Sec. IV-D is

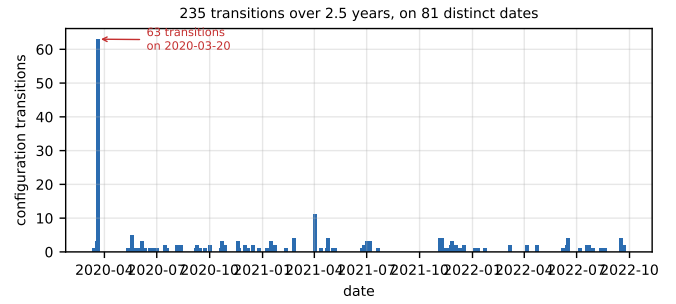


Fig. 10. Configuration transitions over the full record. 2020-03-20 alone accounts for 63.

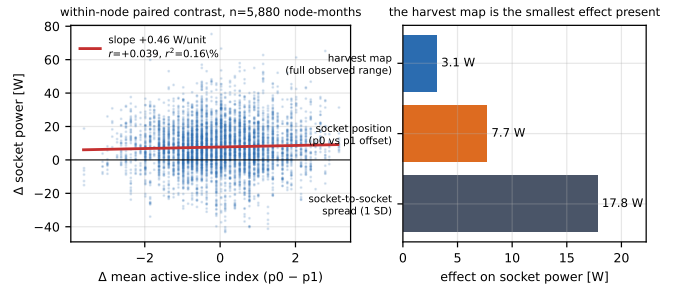


Fig. 11. Operational impact of the harvest map. Left: within-node paired contrast. Right: the harvest map's full observed range moves socket power less than half as much as mere socket position, and a sixth as much as the socket-to-socket spread.

supported. It remains an inference from thermal coupling rather than from a vendor floorplan; an annotated die shot would confirm it directly.

- 3) **[OI-2, *blocking*] A March 2020 BMC firmware update could have remapped sensor IDs**, mimicking reconfiguration; 86 of the 235 transitions fall in the three days 2020-03-19–21. Needs CINECA maintenance logs. Does not affect the boot-time-only conclusion.
- 4) **[OI-4, *blocking*] Five references are unverified** and a novelty check against prior telemetry-based binning inference is outstanding.
- 5) **[OI-3] Defect-driven versus policy-driven harvesting** cannot be separated from one machine; both are reported.
- 6) **[OI-5] Collection dropout may not be random.** Monthly coverage ranges from 18% to 99.4%; if dropout correlates with lot, Sec. IV-G needs a robustness pass.
- 7) **[OI-6, OI-7] both resolved.** The sub-16 socket-days were dominated by an extraction defect on our side, now gated; the nine that survive are the guard episodes of Sec. IV-H. The two nodes appearing to change within a day are same-day reboots.
- 8) **[OI-8] The mechanism behind p0/p1 relabelling is unknown;** any per-socket longitudinal study on this dataset risks silently mixing two dies.
- 9) **Common-mode artefact** in the thermal analysis, addressed by same-distance contrasts. **Sample size mat-**

ters more than we first appreciated: the three-day 2020-04 sample overstated the within/cross-slice contrast threefold and inverted the floorplan conclusion. All thermal results here use a full month and both sockets.

- 10) **[OI-10] Single system**; findings may reflect CINECA procurement rather than POWER9 generally.

VIII. CONCLUSION

Out-of-band telemetry that supercomputer operators already collect inadvertently publishes a manufacturing secret: which units a vendor fused off each individual die. Recovering these maps for 1,962 POWER9 sockets shows harvesting operates exactly at the slice—the unit sharing an L2/L3 block—with no exceptions in 1.67M socket-days; that the maps are per-die rather than per-SKU, with 443 of 495 possible patterns realised; that harvested slices cluster in sensor-index space; and that a sharp changepoint at node 440 separates two procurement populations with different binning statistics. Over 2.5 years the maps are static to within 0.25 percentage points per month, change only across reboots, and exhibit three field core-guard episodes, each removing exactly one slice while every other sensor on the socket keeps reporting.

Two things deserve emphasis. The thermal probe recovers the physical slice ordering from telemetry alone, which is what licenses reading the clustering spatially and connects the measurement to the yield literature; it also shows how much this depends on sample size, since a three-day sample gave the opposite answer. And the harvest map turns out to have no measurable operational effect—a negative result we consider as useful as the positive ones, since it tells operators what they can safely ignore. The broader point is that fleet telemetry is a viable instrument for hardware provenance, and that its unintended disclosures deserve attention from those who publish such datasets.

ARTEFACT AVAILABILITY

All analysis code, the derived per-(node, socket, core, day) tables, and the recovered harvest maps are available at the artefact repository accompanying this paper. Every result is reproducible from the public M100 ExaData Zenodo releases [1]; the derived tables (15 MB) suffice for all structural and longitudinal results without re-downloading the 49.9 TB source.

REFERENCES

- [1] A. Borghesi, C. Di Santi, M. Molan, M. S. Ardebili, A. Mauri, M. Guarrasi, D. Galetti, M. Cestari, F. Barchi, L. Benini, F. Beneventi, and A. Bartolini, “M100 ExaData: a data collection campaign on the CINECA’s Marconi100 Tier-0 supercomputer,” *Scientific Data*, vol. 10, no. 1, p. 288, 2023.
- [2] B. Thompto, “POWER9: Processor for the cognitive era,” in *Hot Chips 28 Symposium (HCS)*. IEEE, 2016, slide 11 documents 24 cores organised as 12 slices, each with 10 MB L3, attached to a 7 TB/s on-chip fabric at 256 GB/s $\times$ 12.
- [3] “POWER9 — microarchitectures — IBM,” WikiChip, 2024, <https://en.wikichip.org/wiki/ibm/microarchitectures/power9>; slice = 2 SMT4 cores + 512 kB L2 + 10 MB L3; die 695 mm², 14 nm.
- [4] J. Stojkovic *et al.*, “HardHarvest: Hardware-supported core harvesting for microservices,” in *Proc. 52nd Annual Int. Symp. on Computer Architecture (ISCA ’25)*, 2025, uses “core harvesting” in the idle-cycle-reclamation sense, distinct from the silicon-binning sense used here.
- [5] IBM, “IBM Power System AC922 data sheet,” IBM Corporation, Tech. Rep., 2019, model 8335-GTG: 2 sockets, 16-core POWER9 at 2.6/3.1 GHz, 4 $\times$ NVIDIA V100.
- [6] I. Baccarelli, “Introduction to the CINECA Marconi100 HPC system,” CINECA HPC user training, 2020, <https://indico.euro-fusion.org/event/341/attachments/293/664/M100.pdf>.
- [7] M100 ExaData contributors, “M100 ExaData reference repository: plugin documentation and rack layout,” [documentation/plugins/ipmi.md](https://github.com/ibm-m100-exadata/documentation/plugins/ipmi.md), [documentation/racks_spatial_distribution.md](https://github.com/ibm-m100-exadata/documentation/racks_spatial_distribution.md), 2023.
- [8] G. Strona, D. Nappo, F. Boccacci, S. Fattorini, and J. San-Miguel-Ayanz, “A fast and unbiased procedure to randomize ecological binary matrices with fixed row and column totals,” *Nature Communications*, vol. 5, p. 4114, 2014.
- [9] I. Koren, Z. Koren, and C. H. Stapper, “A unified negative-binomial distribution for yield analysis of defect-tolerant circuits,” *IEEE Transactions on Computers*, vol. 42, no. 6, pp. 724–737, 1993.
- [10] S. J. Bae, J. Y. Hwang, and W. Kuo, “Yield prediction via spatial modeling of clustered defect counts across a wafer map,” *IIE Transactions*, vol. 39, no. 12, pp. 1073–1083, 2007.
- [11] IBM, “POWER9: Deconfiguring hardware, examples of the reason processor cores were unconfigured, and field core override function overview,” IBM Documentation, 2021, <https://www.ibm.com/docs/en/power9?topic=configuration-deconfiguring-hardware>; covers predictive/functional deconfiguration, Dynamic Processor Deallocation, and field core override (feature code 2319).
- [12] OpenPOWER, “opal-gard: manipulate the GUARD partition on OpenPower hardware,” skiboot, [external/gard/opal-gard.1](https://github.com/open-power/skiboot), 2023, <https://github.com/open-power/skiboot>; GARD records live in a PNOR flash partition and support list, show and clear.
- [13] Y. Inadomi, T. Patki, K. Inoue, M. Aoyagi, B. Rountree, M. Schulz, D. Lowenthal, Y. Wada, K. Fukazawa, M. Ueda, M. Kondo, and I. Miyoshi, “Analyzing and mitigating the impact of manufacturing variability in power-constrained supercomputing,” in *Proc. Int. Conf. for High Performance Computing, Networking, Storage and Analysis (SC ’15)*. ACM, 2015.
- [14] B. Acun, P. Miller, and L. V. Kale, “Variation among processors under Turbo Boost in HPC systems,” in *Proc. 2016 Int. Conf. on Supercomputing (ICS ’16)*. ACM, 2016, pp. 6:1–6:12.
- [15] P. Sinha, A. Guliani, R. Jain, B. Tran, M. D. Sinclair, and S. Venkataraman, “Not all GPUs are created equal: characterizing variability in large-scale, accelerator-rich systems,” in *Proc. Int. Conf. for High Performance Computing, Networking, Storage and Analysis (SC ’22)*, 2022, arXiv:2208.11035.
- [16] “Inherent hardware identifiers: advancing IC traceability and provenance in the multi-die era,” *Journal of Electronic Testing*, 2025.
- [17] “Effective node-level anomaly detection in HPC systems via coarse-grained clustering and fine-grained model sharing,” in *Proc. Int. Conf. for High Performance Computing, Networking, Storage and Analysis (SC ’25)*, 2025.
- [18] “Bad nodes considered harmful: how to find and fix the problem,” in *Lecture Notes in Computer Science*, 2020.
- [19] “Downstream analyses of the M100 ExaData release,” arXiv:2607.22548 (semantic foundation models for HPC telemetry); arXiv:2505.14796 (actionable energy insights); arXiv:2507.06107 (ontology for HPC operational data analytics), 2025.